

CHAPTER 09 The mole

Chemists in fields as diverse as environmental monitoring, pharmaceuticals and fuel production routinely carry out chemical reactions in their work. It is important for them to be able to measure specific quantities of chemicals quickly and easily, in part because the amount of products formed depends on the amount of reactants.

At the end of this chapter, you will have a greater understanding of the way in which chemists measure quantities of chemicals, in particular, the way they can accurately determine the number of particles in samples of elements and compounds simply by weighing them. This is essential for designing and producing materials, including cosmetics, fuels, fertilisers, pharmaceuticals and building materials.

Science understanding

- the mole is a precisely defined quantity of matter equal to Avogadro's number of particles
- the mole concept relates mass, moles and molar mass and, with the Law of Conservation of Mass, can be used to calculate the masses of reactants and products in a chemical reaction
- percentage composition of a compound can be calculated from the relative atomic masses of the elements in the compound and the formula of the compound
- molecular formulae represent the number and type of atoms present in the molecules

WACE Chemistry ATAR Course Year 11 Syllabus © School Curriculum and Standards Authority, Government of Western Australia, 2014; reproduced by permission.

9.1 Masses of particles

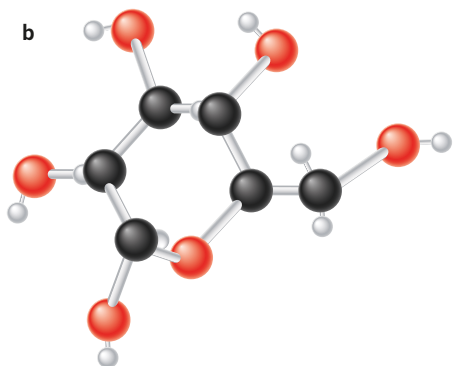


FIGURE 9.1.1 A teaspoon of glucose crystals contains an incredibly large number of extremely small glucose molecules. The single glucose molecule pictured contains 6 carbon atoms (black), 12 hydrogen atoms (white) and 6 oxygen atoms (red) bonded together.

Atoms are particles so small that they cannot be counted or weighed individually or even in groups of thousands or millions. For example, the number of carbon atoms in this dot, ·, printed on this page, is approximately 1×10^{20} atoms. The mass of one atom is incredibly small. For example, one atom of carbon has a mass of approximately 2×10^{-23} g.

Figure 9.1.1 shows a model of a glucose molecule. Glucose is a type of sugar used as an energy source by almost all living organisms on Earth. One glucose molecule contains 6 carbon atoms, 12 hydrogen atoms and 6 oxygen atoms bonded together. One glucose molecule has an actual mass of 3×10^{-22} g. This means that the teaspoon of glucose crystals shown in Figure 5.1.1 contains approximately 1.4×10^{22} glucose molecules.

Such small masses are not easily measured and can be inconvenient to use in calculations. This section will introduce you to the ways scientists determine and use the masses of different particles.

RELATIVE MASSES

The masses used most frequently in chemistry are relative masses, rather than actual masses. The standard to which all masses are compared is the mass of an atom of the common **isotope** of carbon, **carbon-12** or ^{12}C , which is given a mass of exactly 12.

RELATIVE MOLECULAR MASS

Some elements and compounds exist as **molecules**, for example oxygen (O_2), nitrogen (N_2) and carbon dioxide (CO_2). For these substances, a **relative molecular mass** can be determined.

The relative molecular mass, symbol M_r , is equal to the sum of the relative atomic masses of the atoms in the molecule. Remember you can obtain relative atomic masses (or standard atomic weights) of elements from the periodic table.

Worked example 9.1.1

CALCULATING THE RELATIVE MOLECULAR MASS OF MOLECULES

Calculate the relative molecular mass of carbon dioxide (CO_2).	
Thinking	Working
Use the periodic table to find the relative atomic mass for the elements represented in the formula.	$A_r(\text{C}) = 12.01$ $A_r(\text{O}) = 16.00$
Determine the number of atoms of each element present, taking into consideration any brackets in the formula.	$1 \times \text{C atom}$ $2 \times \text{O atoms}$
Determine the relative molecular mass by adding the appropriate relative atomic masses.	$M_r = 1 \times A_r(\text{C}) + 2 \times A_r(\text{O})$ $= 1 \times 12.01 + 2 \times 16.00$ $= 44.01$

Worked example: Try yourself 9.1.1

CALCULATING THE RELATIVE MOLECULAR MASS OF MOLECULES

Calculate the relative molecular mass of nitric acid (HNO_3).

i Recall that isotopes are atoms of the same element that have different numbers of neutrons in their nucleus. So isotopes have the same atomic number but different mass number.

i The relative molecular mass is the mass of one molecule of that substance relative to the mass of a ^{12}C atom taken as 12 units exactly.

RELATIVE FORMULA MASS

Some compounds, such as sodium chloride (NaCl) and magnesium oxide (MgO), do not exist as molecules but rather as ionic lattices. For ionic compounds, the term **relative formula mass** is used. Relative formula mass, like relative molecular mass, is calculated by taking the sum of the relative atomic masses of the elements in the formula.

Worked example 9.1.2

CALCULATING THE RELATIVE FORMULA MASS OF IONIC COMPOUNDS

Calculate the relative formula mass of magnesium hydroxide (Mg(OH) ₂).	
Thinking	Working
Use the periodic table to find the relative atomic mass for the elements represented in the formula.	$A_r(\text{Mg}) = 24.31$ $A_r(\text{O}) = 16.00$ $A_r(\text{H}) = 1.008$
Determine the number of atoms of each element present, taking into consideration any brackets in the formula.	$1 \times \text{Mg atom}$ $1 \times 2 = 2 \text{ O atoms}$ $1 \times 2 = 2 \text{ H atoms}$
Determine the relative formula mass by adding the appropriate relative atomic masses.	Relative formula mass $= 1 \times A_r(\text{Mg}) + 2 \times A_r(\text{O}) + 2 \times A_r(\text{H})$ $= 24.31 + 2 \times 16.00 + 2 \times 1.008$ $= 58.33$

i The relative formula mass is the mass of a formula unit relative to the mass of an atom of ¹²C taken as 12 units exactly. It is numerically equal to the sum of the relative atomic masses of the elements in the formula.

Worked example: Try yourself 9.1.2

CALCULATING THE RELATIVE FORMULA MASS OF IONIC COMPOUNDS

Calculate the relative formula mass of copper(II) nitrate (Cu(NO₃)₂).

i Relative molecular mass and relative formula mass, like relative atomic mass, have no units, because they are based on the masses of atoms of elements compared with the mass of the carbon-12 isotope.

9.1 Review

SUMMARY

- The carbon-12 isotope is assigned a mass of exactly 12 units, and the relative atomic masses of atoms of elements are compared to this.
- The relative molecular mass of molecules, or formula mass of ionic compounds, is calculated from the sum of the relative atomic masses of their constituent elements.

KEY QUESTIONS

- Calculate the relative molecular masses of:
 - sulfuric acid (H₂SO₄)
 - ammonia (NH₃)
 - ethane (C₂H₆).
- Calculate the relative formula mass of:
 - potassium chloride (KCl)
 - sodium carbonate (Na₂CO₃)
 - aluminium sulfate (Al₂(SO₄)₃).

9.2 Introducing the mole

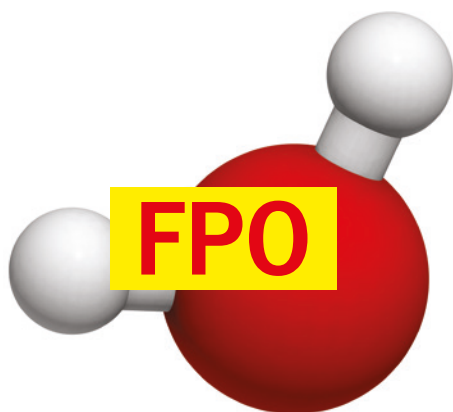


FIGURE 9.2.1 Each of these ice cubes contains more than 10^{23} water (H_2O) molecules. Molecules and atoms are so small, and the numbers of them in everyday samples are so large, that it would be very inconvenient to count them individually.

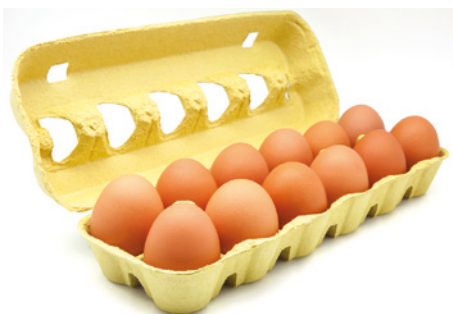


FIGURE 9.2.2 Twelve eggs is one dozen, 24 eggs is two dozen and six eggs is half a dozen.

It is often essential for chemists to be able to measure an exact number of particles of an element or compound. However, the particles in elements and compounds are so small that it would be difficult to count atoms, ions or molecules individually. If it were possible to count individual particles, the numbers in even very small samples would be huge and very inconvenient to work with.

The ice cubes shown in Figure 9.2.1 each contain more than 10^{23} water molecules (H_2O). As each water molecule is composed of two hydrogen atoms and one oxygen atom, the number of individual atoms in each ice cube is greater than 10^{23} . A quantity that allows chemists to measure accurate amounts of extremely small particles is required.

In this section you will learn about the quantity used by chemists; the mole.

THE CHEMIST'S COUNTING UNIT

A dozen is a convenient quantity for buying the eggs shown in Figure 9.2.2. For atoms, ions and molecules that are much smaller than eggs, a quantity that describes a much larger number is needed.

The quantity used by scientists is the **mole**. The mole is often referred to as the 'amount of substance' and is given the symbol n , and the unit, mol.

So, $n(\text{glucose}) = 2 \text{ mol}$ is read as 'the amount of glucose molecules is 2 moles'.

Chemists use the mole as a counting measure. Figure 9.2.3 shows some quantities that you would be very familiar with such as pair, dozen and ream. One dozen is equal to 12 and two dozen equals 24, 20 dozen equals 240 and half a dozen equals 6. In the same way, chemists know that one mole is equivalent to a certain number and that 2 moles, 20 moles and half a mole are all multiples of that number.



FIGURE 9.2.3 Convenient quantities that you would be very familiar with. A pair of shoes equals two, a dozen roses equals 12, a ream of paper equals 500 sheets.

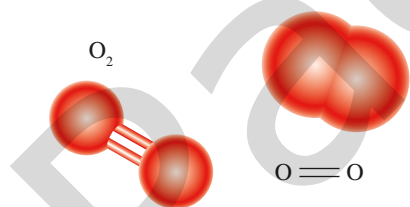


FIGURE 9.2.4 Four different ways chemists represent the oxygen molecule: a formula (O_2), a structural formula ($\text{O}=\text{O}$) and two coloured molecular models. Each formula or model shows that one oxygen molecule contains two oxygen atoms.

Information provided by molecular formulae

When referring to a mole of a substance, it is important to indicate which particle is being specified. Figure 9.2.4 shows four different ways that chemists use to represent the oxygen molecule. The oxygen molecule is most commonly described by the **formula** O_2 .

The expression, 'one mole of oxygen' is ambiguous because it could describe one mole of oxygen atoms (O) or one mole of oxygen molecules (O_2). As there are two atoms in each oxygen molecule, one mole of oxygen molecules will contain two moles of oxygen atoms. As you learnt in Chapter 6, the molecular formula of a substance indicates the number of atoms of each element in one molecule of a substance. For example, in one mole of oxygen gas, there are two moles of oxygen atoms.

So if you know the number of moles of a certain substance, the formula of the substance allows you to also know how many moles of the atoms or ions that make up that substance. Some examples of this are given in Table 9.2.1.

TABLE 9.2.1 Examples of the use of the mole as a measure of the number of particles

Number of moles of substance	Information that can be obtained from the moles of the substance
1 mole of hydrogen gas (H_2)	1 mole of hydrogen molecules (H_2) 2 moles of hydrogen atoms (H)
2 moles of water (H_2O)	2 moles of water molecules (H_2O) 4 moles of hydrogen atoms (H) 2 moles of oxygen atoms (O)
2 moles of calcium chloride (CaCl_2)	2 moles of Ca^{2+} ions 4 moles of Cl^- ions
10 moles of glucose ($\text{C}_6\text{H}_{12}\text{O}_6$)	10 moles of glucose ($\text{C}_6\text{H}_{12}\text{O}_6$) molecules 60 moles of carbon atoms 120 moles of hydrogen atoms 60 moles of oxygen atoms

Worked example 9.2.1

CALCULATING THE NUMBER OF MOLES OF ATOMS GIVEN THE NUMBER OF MOLES OF MOLECULES

Calculate the amount, in mol, of hydrogen atoms in 3.6 mol of sulfuric acid (H_2SO_4).	
Thinking	Working
List the data given in the question next to the appropriate symbol. Include units.	The number of mol of hydrogen atoms is the unknown so: $n(\text{H}) = ?$ $n(\text{H}_2\text{SO}_4) = 3.6 \text{ mol}$
Calculate the amount, in mol, of hydrogen atoms from the amount of sulfuric acid molecules and the molecular formula.	$n(\text{H}) = n(\text{H}_2\text{SO}_4) \times 2$ $= 3.6 \times 2$ $= 7.2 \text{ mol}$

Worked example: Try yourself 9.2.1

CALCULATING THE NUMBER OF MOLES OF ATOMS GIVEN THE NUMBER OF MOLES OF MOLECULES

Calculate the amount, in mol, of hydrogen atoms in 0.75 mol of water (H_2O).

AVOGADRO'S NUMBER

One mole of any substance is defined as the same number of particles as there are atoms in exactly 12 grams of carbon-12. This number of atoms has been experimentally determined to be $6.022\,141 \times 10^{23}$ atoms. That's 602 214 100 000 000 000 000 000 atoms!

This number is commonly rounded to 6.022×10^{23} and is referred to as **Avogadro's number** or Avogadro constant. It is given the symbol N_A .

i $N_A = 6.022 \times 10^{23} \text{ mol}^{-1}$

Therefore, 1 mol of particles contains 6.022×10^{23} particles.

CHEMFILE

Avogadro's number

It is very difficult to imagine just how big Avogadro's number really is, especially when atoms, ions and molecules are so small. Here are some examples to help:

- 6.022×10^{23} grains of sand, placed side by side, would stretch from Earth to the Sun and back about 7 million times.
- A computer counting 10 billion times every second would take 2 million years to reach 6.022×10^{23} .
- 6.022×10^{23} of the marshmallows shown in Figure 9.2.6 would cover Australia to a depth of 900 km!



FIGURE 9.2.6 One mole or 6.022×10^{23} marshmallows would cover Australia to a depth of 900 km!

Avogadro's number is an enormous number, but the extremely small size of atoms, ions and molecules means that one mole of most elements and compounds does not have a large mass or take up a large volume.

For example, in Figure 9.2.5 you can see that one mole of water, that is 6.022×10^{23} water molecules, has a volume of only 18 mL, whereas, one mole of table salt (NaCl), has a mass of 58.5 grams.



FIGURE 9.2.5 One mole of sodium chloride and one mole of water contain the same number of particles, 6.022×10^{23} .

Given that 1 mol of a substance contains 6.022×10^{23} particles, then:

- 2 mol of a substance contains $2 \times (6.022 \times 10^{23}) = 1.204 \times 10^{24}$ particles
- 0.3 mol of a substance contains $0.3 \times (6.022 \times 10^{23}) = 1.81 \times 10^{23}$ particles
- 4.70×10^{23} particles = $\frac{4.70 \times 10^{23}}{6.022 \times 10^{23}} = 0.781$ mol
- 7.35×10^{24} particles = $\frac{7.35 \times 10^{24}}{6.022 \times 10^{23}} = 12.2$ mol.

As you can see from Figure 9.2.7, a mathematical relationship exists between the number of particles, N , and the amount of substance in moles, n . This relationship can be written as:

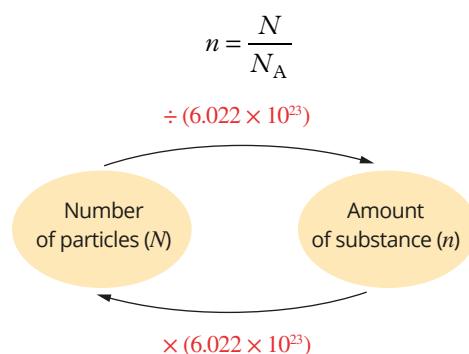


FIGURE 9.2.7 Relationship between number of particles and amount of substance in moles

Calculations using the mole and Avogadro's number

Three quantities have been introduced so far:

- the mole, which is given the symbol n and the unit mol
- Avogadro's number, which is given the symbol N_A and has the value 6.022×10^{23}
- the actual number of particles (atoms, ions or molecules), which is given the symbol N .

The mathematical relationship that links the three quantities is $n = \frac{N}{N_A}$.

The following worked examples show you how to use this relationship in calculations.

Worked example 9.2.2

CALCULATING THE NUMBER OF MOLECULES

Calculate the number of molecules in 3.5 moles of water (H ₂ O).	
Thinking	Working
List the data given in the question next to the appropriate symbol. Include units.	The number of water molecules is the unknown, so: $N(\text{H}_2\text{O}) = ?$ $n(\text{H}_2\text{O}) = 3.5 \text{ mol}$ $N_A = 6.022 \times 10^{23}$
Rearrange the formula to make the unknown the subject.	$n = \frac{N}{N_A}$ so $N(\text{H}_2\text{O}) = n \times N_A$
Substitute in data and solve for the answer.	$N(\text{H}_2\text{O}) = n \times N_A$ $= 3.5 \times 6.022 \times 10^{23}$ $= 2.1 \times 10^{24} \text{ molecules}$

Worked example: Try yourself 9.2.2

CALCULATING THE NUMBER OF MOLECULES

Calculate the number of molecules in 1.6 moles of carbon dioxide (CO₂).

Worked example 9.2.3

CALCULATING THE NUMBER OF ATOMS

Calculate the number of oxygen atoms in 2.5 mol of oxygen gas (O ₂).	
Thinking	Working
List the data given in the question next to the appropriate symbol. Include units.	The number of oxygen atoms is the unknown so: $N(\text{O}) = ?$ $n(\text{O}_2) = 2.5 \text{ mol}$ $N_A = 6.022 \times 10^{23}$
Calculate the amount, in mol, of oxygen atoms from the amount of oxygen molecules and the molecular formula.	$n(\text{O}) = n(\text{O}_2) \times 2$ $= 2.5 \times 2$ $= 5.0 \text{ mol}$
Rearrange the formula to make the unknown the subject.	$n = \frac{N}{N_A}$ so $N(\text{O}) = n \times N_A$
Substitute in data and solve for the answer.	$N(\text{O}) = n \times N_A$ $= 5.0 \times 6.022 \times 10^{23}$ $= 3.0 \times 10^{24} \text{ atoms}$

Worked example: Try yourself 9.2.3

CALCULATING THE NUMBER OF ATOMS

Calculate the number of hydrogen atoms in 0.35 mol of methane (CH_4).

Worked example 9.2.4

CALCULATING THE NUMBER OF MOLES OF PARTICLES GIVEN THE NUMBER OF PARTICLES

Calculate how many moles of ammonia are in 2.5×10^{22} of ammonia molecules.

Thinking	Working
List the data given in the question next to the appropriate symbol. Include units.	The number of mol of ammonia molecules is the unknown so: $n(\text{NH}_3) = ?$ $N(\text{NH}_3) = 2.5 \times 10^{22}$ molecules $N_A = 6.022 \times 10^{23}$
Rearrange the formula to make the unknown the subject.	$n = \frac{N}{N_A}$ n is the unknown so rearrangement not required
Substitute in data and solve for the answer.	$n(\text{NH}_3) = \frac{N}{N_A} = \frac{2.5 \times 10^{22}}{6.022 \times 10^{23}}$ $= 0.042 \text{ mol}$

Worked example: Try yourself 9.2.4

CALCULATING THE NUMBER OF MOLES OF PARTICLES GIVEN THE NUMBER OF MOLES OF PARTICLES

Calculate how many moles of magnesium are in 8.1×10^{20} of magnesium atoms.

9.2 Review

SUMMARY

- A mole is a convenient quantity for counting particles. The mole is given the symbol n and the unit mol.
- One mole is defined as the amount of substance that contains the same number of 'specified' particles as there are atoms in 12 g of carbon-12.
- The number of particles in 1 mol is given the symbol N_A . This is known as Avogadro's number and has the numerical value of 6.022×10^{23} .
- The formula $n = \frac{N}{N_A}$ can be used or rearranged to calculate the amount or number of specified particles in a sample.

KEY QUESTIONS

- 1 Calculate the number of:
 - a atoms in 2.0 mol of sodium atoms (Na)
 - b molecules in 0.10 mol of nitrogen molecules (N_2)
 - c atoms in 20.0 mol of carbon atoms (C)
 - d molecules in 4.2 mol of water molecules (H_2O)
 - e atoms in 1.0×10^{-2} mol of iron atoms (Fe)
 - f molecules in 4.62×10^{-5} mol of CO_2 molecules.
- 2 Calculate the number of moles in:
 - a 3.0×10^{23} molecules of water (H_2O)
 - b 1.5×10^{23} atoms of neon (Ne)
 - c 4.2×10^{25} atoms of iron (Fe)
 - d 4.2×10^{25} molecules of ethanol (C_2H_5OH).
- 3 Calculate the amount, in mol, of:
 - a sodium atoms in 1.0×10^{20} of sodium atoms
 - b aluminium atoms in 1.0×10^{20} of aluminium atoms
 - c chlorine molecules in 1.0×10^{20} of chlorine molecules.
- 4 Calculate the amount, in mol, of:
 - a chlorine atoms in 0.4 mol of chlorine (Cl_2)
 - b hydrogen atoms in 1.2 mol of methane (CH_4)
 - c hydrogen atoms in 0.12 mol of ethane (C_2H_6)
 - d oxygen atoms in 1.5 mol of sodium sulfate (Na_2SO_4).



FIGURE 9.3.1 A digital balance is a piece of laboratory equipment used for weighing.

9.3 Molar mass

Chemical laboratories often contain a balance like the one in Figure 9.3.1, which is used for weighing. If a chemist knows that a specific mass of a substance always contains a specific number of particles, it is possible to easily weigh a sample of the substance and calculate the exact number of particles present in the sample.

In this section you will learn about how the amount of a substance, measured in moles, is related to the mass of the substance.

MOLAR MASS

Chemists have defined the mole so that the number of moles of a substance can be determined by simply measuring its mass.

The particles of different elements and compounds have different masses. Therefore, the masses of one mole of different elements or compounds will also be different. This is like saying that the mass of one dozen oranges will be greater than the mass of one dozen mandarins because one orange is heavier than one mandarin. The mass, in grams, of one mole of a particular element or compound is known as its **molar mass**. It is given the symbol, M , and the unit, g mol^{-1} .

Remember that a mole is defined as the amount of substance that contains the same number of specified particles as there are atoms in 12.00 g of carbon-12.

- 1 atom of ^{12}C has a **relative atomic mass** of 12 exactly.
- 1 mole of atoms of ^{12}C has a **mass** of 12 *grams* exactly.

Naturally occurring carbon is mainly composed of the ^{12}C isotope, but because there is a small amount of the ^{13}C isotope also present the molar mass of carbon is 12.01 g mol^{-1} .

Consider an atom of ^{12}C and an atom of ^{24}Mg . Carbon-12 has been assigned a relative isotopic mass of 12 exactly. On that scale, the relative isotopic mass of ^{24}Mg is 24. Since one mole of ^{12}C atoms weighs exactly 12 g, then one mole of ^{24}Mg must weigh twice as much, 24 g.

Table 9.3.1 shows you how to calculate the molar masses of some common substances.

TABLE 9.3.1 Calculating the molar mass of a substance by adding the relative atomic masses for each atom present in the substance based on the molecular or ionic formula

Substance	Relative atomic masses	Molar mass of substance
Na	Na: 22.99	$= 22.99 \text{ g mol}^{-1}$
O_2	O: 16.00	$= 2 \times 16.00$ $= 32.00 \text{ g mol}^{-1}$
H_2O	H: 1.008 O: 16.00	$= (2 \times 1.008) + 16.00$ $= 18.016 \text{ g mol}^{-1}$
CO_2	C: 12.01 O: 16.00	$= 12.01 + (2 \times 16.00)$ $= 44.01 \text{ g mol}^{-1}$
NaNO_3	Na: 22.99 N: 14.01 O: 16.00	$= 22.99 + 14.01 + (3 \times 16.00)$ $= 85.00 \text{ g mol}^{-1}$

From the calculations in Table 9.3.1 and the photograph of one mole of some common substances in Figure 9.3.2, you can see that one mole of each substance has a different mass.

- i**
- The molar mass of an element is the mass of one mole of the element. It is equal to the relative atomic mass of the element expressed in grams.
 - The molar mass of a compound is the mass of one mole of the compound. It is equal to the relative molecular mass or relative formula mass of the compound expressed in grams.
 - The molar mass is given the symbol, M , and the unit, g mol^{-1} .



FIGURE 9.3.2 One mole of each substance has a different mass.

Counting by weighing

A useful relationship links the amount of a substance (n), in mol, its molar mass (M), in g mol^{-1} , and the given mass of the substance (m), in grams.

Mass of a given amount of substance (g) = amount of substance (mol) $\times$ molar mass (g mol^{-1}).

This can be written as $m = n \times M$ and rearranged to the following relationship.

i The relationship between the amount of substance, n , its molar mass, M , and the given mass of a substance, m , is:

$$\text{Amount in mol} \longrightarrow n = \frac{m}{M}$$

Mass in g
↙
↘
Molar mass in g mol^{-1}

Worked example 9.3.1

CALCULATING THE MASS OF A SUBSTANCE

Calculate the mass of 0.35 mol of magnesium nitrate ($\text{Mg}(\text{NO}_3)_2$).	
Thinking	Working
List the data given to you in the question. Remember that whenever you are given a chemical formula, you can calculate the molar mass.	$m(\text{Mg}(\text{NO}_3)_2) = ? \text{ g}$ $n(\text{Mg}(\text{NO}_3)_2) = 0.35 \text{ mol}$ $M(\text{Mg}(\text{NO}_3)_2) = 24.31 + (2 \times 14.01) + (6 \times 16.00)$ $= 148.33 \text{ g mol}^{-1}$
Calculate the mass of magnesium nitrate using: $n = \frac{m}{M}$	$n = \frac{m}{M}$ so $m = n \times M$ $m(\text{Mg}(\text{NO}_3)_2) = 0.35 \times 148.33$ $= 52 \text{ g}$

Worked example: Try yourself 9.3.1

CALCULATING THE MASS OF A SUBSTANCE

Calculate the mass of 4.68 mol of sodium carbonate (Na_2CO_3).

You are now in a position to count atoms by weighing. When you use the mole, you are effectively counting the number of particles in a substance. The total number of particles present in a substance is equal to the number of moles of the substance multiplied by 6.022×10^{23} .

Some calculations require you to use both of the formulae $n = \frac{m}{M}$ and $n = \frac{N}{N_A}$.
Worked Example 9.3.2 is such a calculation.

Worked example 9.3.2

CALCULATING THE NUMBER OF MOLECULES

Calculate the number of CO ₂ molecules present in 22.0g of carbon dioxide.	
Thinking	Working
List the data given to you in the question. Convert mass to grams if required. Remember that whenever you are given a formula you can calculate the molar mass.	$N(\text{CO}_2) = ?$ $M(\text{CO}_2) = 12.0 + (2 \times 16.0) = 44.01 \text{ g mol}^{-1}$ $m(\text{CO}_2) = 22.0 \text{ g}$
Calculate the amount, in mol, of CO ₂ using: $n = \frac{m}{M}$	$n(\text{CO}_2) = \frac{m}{M}$ $= \frac{22.0}{44.01}$ $= 0.4999 \text{ mol}$
Calculate the number of CO ₂ molecules using: $n = \frac{N}{N_A}$	$n = \frac{N}{N_A}$ so $N = n \times N_A$ $N(\text{CO}_2) = 0.4999 \times 6.022 \times 10^{23}$ $= 3.01 \times 10^{23} \text{ molecules}$

Worked example: Try yourself 9.3.2

CALCULATING THE NUMBER OF MOLECULES

Calculate the number of sucrose molecules in a teaspoon (4.2 g) of sucrose (C₁₂H₂₂O₁₁).

CHEMISTRY IN ACTION

The sting of a bee

The formula for pentyl ethanoate, C₇H₁₄O₂, is represented by the structure shown in Figure 9.3.3.

Pentyl ethanoate (C₇H₁₄O₂) is the compound that gives bananas their characteristic odour. It is also a pheromone released by bees when they sting. A pheromone is a chemical produced by an animal or insect that changes the behaviour of another animal or insect of the same species. Bees release pentyl ethanoate as an alarm pheromone to alert other bees to danger. The compound is released near the sting shaft and attracts other bees to the area, where the group behaves defensively, charging and

stinging. Smoke masks the alarm pheromone, interrupting the defensive response and calming the bees, allowing beekeepers an opportunity to work with the beehive.

Each time a bee stings, one-thousandth of a milligram (1.0 × 10⁻⁶g) of pentyl ethanoate is released. Knowing the mass of pentyl ethanoate released in a bee sting enables chemists to calculate the number of pentyl ethanoate molecules in each sting using the relationship between mass and mole.

As the molar mass of pentyl ethanoate (C₇H₁₄O₂) is equal to 130.2 g mol⁻¹, using the formula $n = \frac{m}{M}$, 1.0 × 10⁻⁶g of pentyl ethanoate is equal to 7.7 × 10⁻⁹mol.

From section 9.2, you know that the number of particles can

be calculated using the formula $N = n \times N_A$. The number of pentyl ethanoate molecules in each bee sting is therefore equal to 4.6 × 10¹⁵.

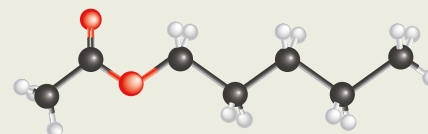


FIGURE 9.3.3 Pentyl ethanoate is a compound released when bees sting.

9.3 Review

SUMMARY

- The molar mass of an element or compound is the mass, in grams, of one mole of that element or compound. Molar mass is given the symbol M and the unit g mol^{-1} .
- The molar mass of an element or compound has the same numerical value as the relative mass of the element or compound.
- The formula $n = \frac{m}{M}$ can be used or rearranged to calculate the mass, amount or molar mass of an element or compound.

KEY QUESTIONS

- 1 Calculate the molar mass of the following, to one decimal place.
 - a nitrogen (N_2)
 - b ammonia (NH_3)
 - c sulfuric acid (H_2SO_4)
 - d iron(III) nitrate ($\text{Fe}(\text{NO}_3)_3$)
 - e ethanoic acid (CH_3COOH)
 - f sulfur atoms (S)
 - g vitamin C (ascorbic acid, $\text{C}_6\text{H}_8\text{O}_6$)
 - h hydrated copper(II) sulfate ($\text{CuSO}_4 \cdot 5\text{H}_2\text{O}$)
- 2 Calculate the mass of the following, to three significant figures.
 - a 1.00 mol of sodium atoms (Na)
 - b 2.00 mol of oxygen molecules (O_2)
 - c 0.100 mol of methane molecules (CH_4)
 - d 0.250 mol of aluminium oxide (Al_2O_3)
- 3 Calculate the amount of the following, in mol, to the appropriate number of significant figures.
 - a H atoms in 5.0 g of hydrogen (H)
 - b H molecules in 5.0 g of hydrogen (H_2)
 - c Al atoms in 2.7 g of aluminium (Al)
 - d CH_4 molecules in 0.4 g of methane (CH_4)
 - e O_2 molecules in 0.10 g of oxygen (O_2)
 - f O atoms in 0.10 g of oxygen (O_2)
 - g P_4 molecules in 1.2×10^{-3} g of phosphorus (P_4)
 - h P atoms in 1.2×10^{-3} g of phosphorus (P_4)
- 4 Calculate the number of atoms in the following, to two significant figures.
 - a 23 g of sodium (Na)
 - b 4.0 g of argon (Ar)
 - c 0.243 g of magnesium (Mg)
 - d 10.0 g of gold (Au)
- 5 Calculate the number of molecules in:
 - i 16 g of oxygen (O_2)
 - ii 2.8 g of nitrogen (N_2).
 - b Calculate the number of oxygen atoms in 3.2 g of sulfur dioxide (SO_2).
 - c Calculate the total number of atoms in 288 g of ammonia (NH_3).

9.4 Percentage composition

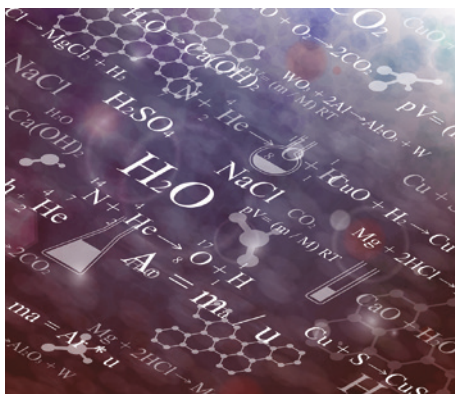


FIGURE 9.4.1 Chemical formulae are part of the language of chemistry.

CHEMFILE

Elemental analysis of magnetite

If the chemical formula of a compound is unknown, the percentage composition by mass may be found using **elemental analysis** in a laboratory. An example is the analysis of the mineral magnetite, one of the three most commonly occurring oxides of iron. The most magnetic of all naturally occurring minerals on Earth, magnetite was used by the Chinese to make the first magnetic compass.

Found in many different types of igneous and metamorphic rocks, large deposits are located in the Pilbara region of Western Australia. Elemental analysis allowed chemists to determine the percentage composition of magnetite as 72.4% iron and 27.6% oxygen (Figure 9.4.3).



FIGURE 9.4.3 Elemental analysis of the mineral magnetite has allowed chemists to determine the percentage composition to be 72.4 % iron and 27.6% oxygen.

Compounds are substances that contain two or more different elements. The relative proportions of each element in a compound can be expressed as a formula, showing either the ratio of atoms of each element in a compound or the actual numbers of atoms in a molecule.

The formulae of compounds are an important part of the language of chemistry. You would be familiar with some of the formulae shown in Figure 9.4.1. formulae are used to represent chemicals in equations and on numerous other occasions.

In this section you will learn how to use the formula of a compound to calculate percentage composition, that is, the percentage of each element in terms of mass.

PERCENTAGE COMPOSITION

The **percentage composition** of a given compound tells you the proportion by mass of the different elements in that compound. The proportion of each element is expressed as a percentage of the total mass of the compound (Figure 9.4.2).

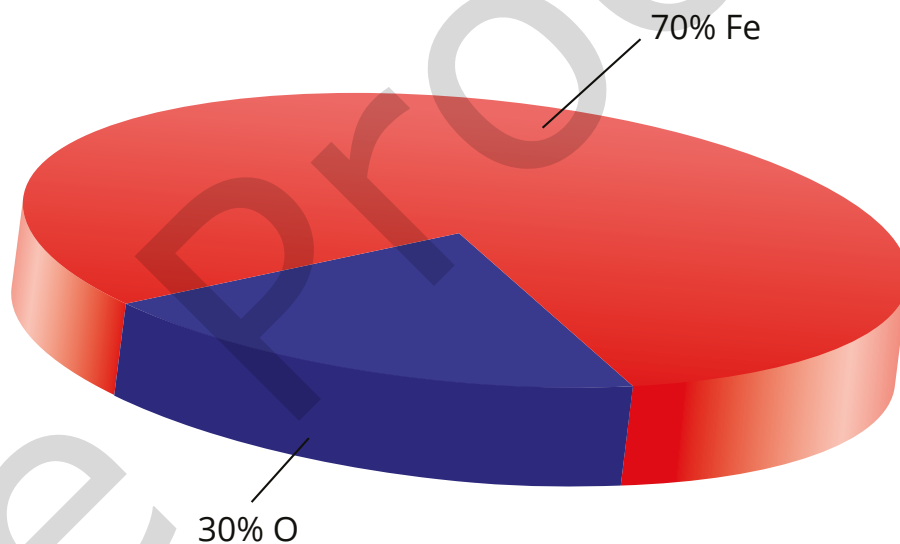


FIGURE 9.4.2 The pie chart shows the percentage composition, by mass, of iron(III) oxide (Fe_2O_3).

Being able to determine percentages by mass is important in chemistry. For example, the iron ore mined in Western Australia is iron(III) oxide, and has the formula Fe_2O_3 . A company that is producing iron from this iron ore will want to know the mass of iron that can be extracted from a given quantity of iron ore. The pie chart in Figure 9.4.2 shows that the percentage of iron in iron(III) oxide is 70%. This means that the company can extract a maximum of 70 g iron per 100 g of iron ore.

If the chemical formula of a compound is known, the percentage composition can be determined from the molar masses of the elements and compound using the following formula.

$$\begin{aligned} & \text{i} \quad \% \text{ by mass of an element in a compound} \\ &= \frac{\text{mass of the element in 1 mol of the compound}}{\text{molar mass of the compound}} \times 100 \end{aligned}$$

Worked example 9.4.1

CALCULATING PERCENTAGE COMPOSITION

Calculate the percentage by mass of aluminium in alumina (Al_2O_3).	
Thinking	Working
Find the molar mass of the compound.	$M(\text{Al}_2\text{O}_3) = (2 \times 26.98) + (3 \times 16.00)$ $= 101.96 \text{ g mol}^{-1}$
Find the total mass of the element in one mole of the compound.	mass of Al in 1 mol $= 2 \times M(\text{Al})$ $= 2 \times 26.98$ $= 53.96 \text{ g}$
Find the percentage by mass of the element in the compound.	% by mass of Al in Al_2O_3 $= \frac{\text{mass of Al in 1 mol of } \text{Al}_2\text{O}_3}{\text{molar mass of } \text{Al}_2\text{O}_3} \times 100$ $= \frac{53.96}{101.96} \times 100$ $= 52.92\%$

Worked example: Try yourself 9.4.1

CALCULATING PERCENTAGE COMPOSITION

Calculate the percentage by mass of nitrogen in ammonium nitrate (NH_4NO_3).

9.4 Review

SUMMARY

- The percentage, by mass, of an element in a compound can be calculated from the mass of the element in 1 mole of the compound and the molar mass of the compound.

KEY QUESTIONS

- Calculate the percentage by mass of:
 - iron in iron(III) oxide (Fe_2O_3)
 - uranium in uranium oxide (U_3O_8)
 - nitrogen in ammonium chloride (NH_4Cl)
 - oxygen in copper(II) nitrate ($\text{Cu}(\text{NO}_3)_2$).
- Nitrogen is an essential element required for healthy plant growth. Each of the following compounds is a fertiliser used by farmers and market gardeners. By calculating the percentage by mass of nitrogen in each, determine the fertiliser that has the highest nitrogen content.
 - urea (NH_2CONH_2)
 - ammonium nitrate (NH_4NO_3)
 - ammonium sulfate ($(\text{NH}_4)_2\text{SO}_4$)
- Calculate the percentage by mass of each element in the following compounds.
 - silicon carbide (SiC)
 - water (H_2O)
 - potassium phosphate (K_3PO_4)
 - sucrose ($\text{C}_{12}\text{H}_{22}\text{O}_{11}$)
 - gold sulfate ($\text{Au}_2(\text{SO}_4)_3$)

Chapter review

KEY TERMS

amount	molar mass
Avogadro's number	mole
carbon-12	percentage composition
elemental analysis	relative formula mass
formula	relative molecular mass

Masses of particles

- Determine the relative molecular mass (M_r) of:
 - water (H_2O)
 - white phosphorus (P_4)
 - carbon monoxide (CO).
- Determine the relative formula mass of:
 - zinc bromide (ZnBr_2)
 - barium hydroxide ($\text{Ba}(\text{OH})_2$)
 - iron(III) carbonate ($\text{Fe}_2(\text{CO}_3)_3$).

Introducing the mole

- For each of the following numbers of molecules, calculate the amount of substance, in mol.
 - 4.50×10^{23} molecules of water (H_2O)
 - 9.00×10^{24} molecules of methane (CH_4)
 - 2.3×10^{28} molecules of chlorine (Cl_2)
 - 1 molecule of sucrose ($\text{C}_{12}\text{H}_{22}\text{O}_{11}$)
- For each of the following amounts of molecular substances, calculate the:
 - number of molecules
 - total number of atoms.
 - 1.45 mol of ammonia (NH_3)
 - 0.576 mol of hydrogen sulfide (H_2S)
 - 0.0153 mol of hydrogen nitrate (HNO_3)
 - 2.5 mol of sucrose ($\text{C}_{12}\text{H}_{22}\text{O}_{11}$)

Molar mass

- How would the molar mass (M) of a compound differ from its relative molecular mass (M_r)?
- What is the molar mass (M) of each of the following?
 - iron (Fe)
 - sulfuric acid (H_2SO_4)
 - sodium oxide (Na_2O)
 - zinc nitrate ($\text{Zn}(\text{NO}_3)_2$)
 - glycine ($\text{H}_2\text{NCH}_2\text{COOH}$)
 - aluminium sulfate ($\text{Al}_2(\text{SO}_4)_3$)
 - hydrated iron(III) chloride ($\text{FeCl}_3 \cdot 6\text{H}_2\text{O}$)
- What is the mass of each of the following?
 - 0.060 mol of ethane (C_2H_6)
 - 0.32 mol of glucose ($\text{C}_6\text{H}_{12}\text{O}_6$)
 - 6.8×10^{-3} mol of urea ($(\text{NH}_2)_2\text{CO}$)
 - 6.12 mol of copper atoms (Cu)

- What is the amount, in mol, of each of the following?
 - carbon atoms in 1.201 g carbon
 - sulfur molecules (S_8) in 10.0 g sulfur
 - methane molecules (CH_4) in 20.0 g methane
 - aspirin molecules ($\text{C}_6\text{H}_4(\text{OCOCH}_3)\text{COOH}$) in 300 mg aspirin
 - aluminium oxide (Al_2O_3) in 3.5 tonnes of aluminium oxide (1 tonne = 1000 kg)
- If 6.022×10^{23} atoms of calcium have a mass of 40.1 g, what is the mass of one calcium atom?
 - If 1.0 mol of water molecules has a mass of 18.0 g, what is the mass of one water molecule?
 - What is the mass of one molecule of carbon dioxide?
- For each of the following molecular substances, calculate the:
 - amount of substance in moles
 - number of molecules
 - total number of atoms.
 - 4.2 g of phosphorus (P_4)
 - 75.0 g of sulfur (S_8)
 - 0.32 g of hydrogen chloride (HCl)
 - 2.2×10^{-2} g of glucose ($\text{C}_6\text{H}_{12}\text{O}_6$)
- What mass of iron (Fe) would contain as many iron atoms as there are molecules in 20.0 g water (H_2O)?
- For each of the following ionic substances calculate the amount of:
 - substance, in moles
 - each ion, in moles.
 - 5.85 g of NaCl
 - 45.0 g of CaCl_2
 - 1.68 g of $\text{Fe}_2(\text{SO}_4)_3$
- If 0.50 mol of a substance has a mass of 72.0 g, what is the mass of 1.0 mol of the substance?
 - If 6.0×10^{22} molecules of a substance have a mass of 10.0 g, what is the molar mass of the substance?
- Calculate the molar mass of a substance if:
 - 2.0 mol of the substance has a mass of 80.0 g
 - 0.10 mol of the substance has a mass of 9.8 g
 - 1.7 mol of the substance has a mass of 74.8 g
 - 3.50 mol of the substance has a mass of 371 g.

- 15 Which of the following metal samples has the greatest mass?

A 100 g copper
B 4.0 mol of iron atoms
C 1.2×10^{24} atoms of silver

- 16 A new antibiotic has been isolated and only 2.0 mg is available. The molar mass is found to be 12.5 kg mol^{-1} .
- Express the molar mass in g mol^{-1} .
 - Calculate the amount of antibiotic, in mol.
 - How many molecules of antibiotic have been isolated?

Percentage composition

- 17 Calculate the percentage by mass of each element in:

a Al_2O_3
b $\text{Cu}(\text{OH})_2$
c $\text{MgCl}_2 \cdot 6\text{H}_2\text{O}$
d $\text{Fe}_2(\text{SO}_4)_3$
e perchloric acid (HClO_4).

- 18 Determine the percentage by mass of carbon in:

a naphthalene (C_{10}H_8)
b ethanoic acid (CH_3COOH)
c urea (NH_2CONH_2)
d aspirin ($\text{C}_6\text{H}_4(\text{OCOCH}_3)\text{COOH}$).

Connecting the main ideas

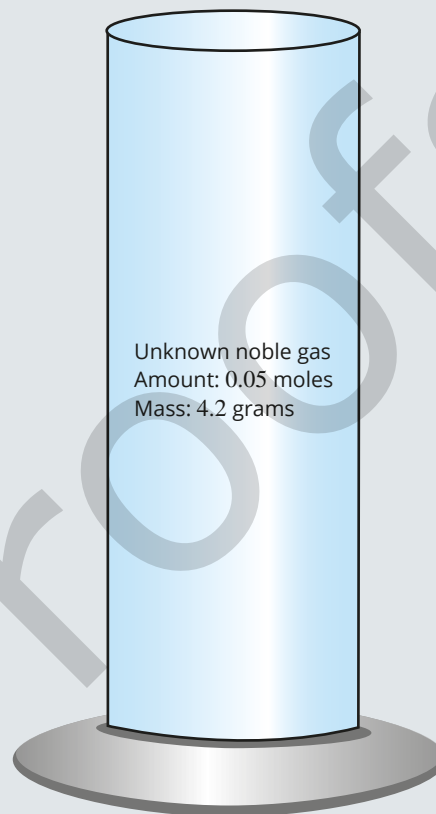
- 19 a How many molecules of water are there in 5.0 moles of ice?
b How many moles of Cu atoms in 2.4×10^{24} atoms of copper?
c What is the mass of a magnesium atom?
d What is the mass of a water molecule? (Molar mass of H_2O is 18.02 g.)
- 20 a The formula for gold sulfate is $\text{Au}_2(\text{SO}_4)_3$. For every 1 mole of $\text{Au}_2(\text{SO}_4)_3$, how many moles are there of:
i gold atoms?
ii sulfur atoms?
iii oxygen atoms?

- b What is the molar mass, M , of gold sulfate?

Use your answer from part b to answer the following. Show your working out.

- c Calculate the number of moles of gold sulfate in 27.5 g of $\text{Au}_2(\text{SO}_4)_3$.
d What is the mass of 0.660 mole of $\text{Au}_2(\text{SO}_4)_3$?
e What is the percentage of gold in gold sulfate?
- 21 The following questions are based on the hydrated salt, copper(II) nitrate hexahydrate, $\text{Cu}(\text{NO}_3)_2 \cdot 6\text{H}_2\text{O}$.
- What is the molar mass, M , of $\text{Cu}(\text{NO}_3)_2 \cdot 6\text{H}_2\text{O}$?
 - 30.0 g of the crystals are weighed out. How many moles of the hydrated salt is this?
 - What is the percentage of water in copper(II) nitrate hexahydrate?

- 22 An analytical chemist studying the noble gases, He, Ne, Ar, Kr, Xe and Rn (group 18 in the periodic table) is sent a sealed glass cylinder with the following label:



Which noble gas was the unknown gas? (Show your reasoning and calculation.)

- 23 As the science of chemistry progressed, chemists such as Avogadro, Lavoisier and Dalton, used a fairly simple (compared to modern electronic methods) way of measuring atomic masses. They would take equal volumes of gases, weigh them, and compare their masses to a standard atomic mass such as oxygen. The formula was:

$$\frac{\text{mass of oxygen}}{\text{atomic mass of oxygen}} = \frac{\text{mass of unknown gas, X}}{\text{atomic mass of unknown gas, X}}$$

During an experiment carried out in 1809 to determine the atomic mass of X, the following data was obtained by Dalton in his laboratory:

Mass of oxygen = 1.0 g
Atomic mass of oxygen = 16.0 g
Mass of unknown gas, X = 2.2 g

- Calculate the atomic mass of X.
- Suggest one reason why this technique may give incorrect results. (Hint: think about the formulae of some gases that you have encountered.)

FPO